

Brown phases in game semantics: a canonical two-bit split and a divisibility obstruction

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Abstract

The Brown invariant of a nonsingular $\mathbb{Z}/4$ -valued quadratic form is the eighth-root phase of a four-class Gauss census. We determine exactly what a game-theoretic reading of that census adds to the characteristic-two Arf problem. Every Brown form has a canonical decomposition

$$q = \widehat{\ell} + 2Q,$$

where $\ell = q \bmod 2$ is linear and Q is an ordinary $\mathbb{F}_2$ -quadratic form. Consequently the four residue classes are precisely the joint census of one linear game bit and the ordinary quadratic bit already isolated by the **tis** problem; the Brown phase introduces no second nonlinear game-semantic problem. We also prove a complementary global no-go: an ambient-coherent Brown refinement of the additive group of all short games is identically zero, because that group is 2-divisible. Finally, the cyclic extension $C_4 \rightarrow C_8 \rightarrow C_2$ shows that a bare central extension by C_4 does not determine a Brown form: a section or phase framing is essential. Together these results dispose of the independent Brown invariant/output question under an explicit synchronized-charge contract. The later weighted-source theorem supplies the ordinary quadratic normal-play bit; what is not claimed here is that a synchronized pair of binary games is itself one canonical four-class normal-, misère-, or loopy-play outcome.

1 The exact question

A Brown form on an $\mathbb{F}_2$ -space V is a function

$$q : V \longrightarrow \mathbb{Z}/4$$

with $q(0) = 0$ and

$$q(x + y) = q(x) + q(y) + 2b(x, y), \tag{1}$$

where $b : V \times V \rightarrow \mathbb{F}_2$ is symmetric and bilinear. Substituting $y = x$ shows

$$b(x, x) = q(x) \bmod 2. \tag{2}$$

Thus b need not be alternating. When b is nonsingular, the classical Brown invariant $\beta(q) \in \mathbb{Z}/8$ is defined by the Gauss sum

$$\sum_{x \in V} i^{q(x)} = 2^{\dim(V)/2} \zeta_8^{\beta(q)}. \tag{3}$$

This is standard mathematics due to Brown and Wall [1, 3]; the repository's **brown_f2** implementation also records the singular radical boundary.

The phrase “game-theoretic reading” has three logically different senses:

- (i) a four-class charge or outcome attached to positions in one selected finite $\mathbb{F}_2$ -space;
- (ii) a quadratic colour attached intrinsically to short-game *values* and compatible with disjunctive sum and ambient enlargement; or
- (iii) a framed evaluator which is handed q and reports its residue.

Without separating these, four raw labels prove nothing: partizan normal play, normal/misere pairs, loopy starter pairs, and scoring games all provide at least four labels. The results below reduce (i) algebraically to a synchronized linear/quadratic charge pair, rule out (ii) nontrivially, and identify (iii) as tautological rather than a new bridge.

2 The canonical two-bit decomposition

Write $\rho : \mathbb{Z}/4 \rightarrow \mathbb{F}_2$ for reduction modulo two and $\widehat{\cdot} : \mathbb{F}_2 \rightarrow \mathbb{Z}/4$ for the distinguished set-theoretic section $0 \mapsto 0, 1 \mapsto 1$. It obeys

$$\widehat{a + c} = \widehat{a} + \widehat{c} + 2ac \quad (a, c \in \mathbb{F}_2), \quad (4)$$

where the left addition is in $\mathbb{F}_2$ and the right addition is in $\mathbb{Z}/4$.

Theorem 1 (canonical Brown split). **[PROVED]** *Let (q, b) satisfy (1). There are unique maps*

$$\ell : V \rightarrow \mathbb{F}_2, \quad Q : V \rightarrow \mathbb{F}_2$$

such that

$$q(x) = \widehat{\ell(x)} + 2Q(x). \quad (5)$$

They have the following properties.

- (i) $\ell = \rho \circ q$ is linear.
- (ii) Q is an ordinary characteristic-two quadratic form with alternating polar form

$$B_Q(x, y) = b(x, y) + \ell(x)\ell(y). \quad (6)$$

- (iii) *Conversely, every linear ℓ and every ordinary $\mathbb{F}_2$ -quadratic form Q determine a unique Brown form by (5); its symmetric polar form is*

$$b = B_Q + \ell \otimes \ell.$$

Hence Brown forms are canonically equivalent to pairs (ℓ, Q) .

Proof. Reducing (1) modulo two kills $2b$, so $\ell = \rho q$ is additive and therefore $\mathbb{F}_2$ -linear. Each residue $q(x)$ has a unique expression (5), which defines Q and proves uniqueness.

Apply (4) to $\ell(x + y) = \ell(x) + \ell(y)$ and compare the two expressions for $q(x + y)$. Cancelling the distinguished lifts and dividing the remaining even equality by two gives

$$Q(x + y) = Q(x) + Q(y) + b(x, y) + \ell(x)\ell(y).$$

Thus the displayed B_Q is the polar form of Q . Equation (2) and $\ell(x)^2 = \ell(x)$ give

$$B_Q(x, x) = b(x, x) + \ell(x)^2 = \ell(x) + \ell(x) = 0,$$

so it is alternating; bilinearity and symmetry follow from those of b and $\ell \otimes \ell$.

Conversely, let ℓ be linear and Q have alternating polar B_Q . Set $q = \widehat{\ell} + 2Q$ and $b = B_Q + \ell \otimes \ell$. Equation (4) and the polarization law for Q give (1); moreover $b(x, x) = \ell(x) = q(x) \bmod 2$. The two constructions are inverse by the uniqueness of the two binary digits of every element of $\mathbb{Z}/4$. $\square$

Corollary 2 (the four-class census is a joint binary census). **[PROVED]** *Under Theorem 1, the four Brown classes are*

$q(x)$	$\ell(x)$	$Q(x)$	$i^{q(x)}$
0	0	0	+1
1	1	0	+i
2	0	1	-1
3	1	1	-i

In particular

$$\sum_x i^{q(x)} = \sum_x i^{\ell(x)} (-1)^{Q(x)}. \quad (7)$$

If $W(R) = \sum_x (-1)^{R(x)}$, then more explicitly

$$\sum_x i^{q(x)} = \frac{1+i}{2} W(Q) + \frac{1-i}{2} W(Q+\ell). \quad (8)$$

When the full Gauss sum is nonzero—in particular when b is nonsingular— β is the octant of the joint (ℓ, Q) imbalance, or equivalently of the correlated pair of ordinary quadratic signed censuses Q and $Q+\ell$. It is not generally the product of a marginal linear census and the Arf bias of Q . On the doubled nonsingular slice $\ell = 0$, the two Walsh sums agree and this reduces to the shipped identity $\beta(2Q) = 4 \operatorname{Arf}(Q)$. For singular forms the same census identity holds, while the phase belongs to the nonsingular core and the radical data is recorded separately.

Proof. The table and (7) follow immediately from $q = \widehat{\ell} + 2Q$. The identity

$$i^\ell = \frac{1+i}{2} + \frac{1-i}{2} (-1)^\ell$$

gives (8). For $\ell = 0$, the Gauss sum is the ordinary real quadratic Gauss sum, whose sign is $(-1)^{\operatorname{Arf}(Q)}$. $\square$

2.1 The Brown phase in ordinary quadratic coordinates

The split gives more than a four-colour relabelling: it computes every Brown octant from ordinary characteristic-two data.

Theorem 3 (phase formulas). **[PROVED]** *Suppose the symmetric Brown polar form b is nondegenerate and put $n = \dim V$.*

(i) *If n is even, B_Q is nondegenerate. Let a be the unique vector satisfying $\ell(x) = B_Q(a, x)$ for every x . Then*

$$\beta(q) = 4 \operatorname{Arf}(Q) + 2Q(a) \pmod{8}. \quad (9)$$

(ii) *If n is odd, $\operatorname{rad} B_Q = \langle w \rangle$ is one-dimensional, $\ell(w) = 1$, and B_Q restricts nondegenerately to $H = \ker \ell$. Then*

$$\beta(q) = 4 \operatorname{Arf}(Q|_H) + \begin{cases} 1, & Q(w) = 0, \\ 7, & Q(w) = 1 \end{cases} \pmod{8}. \quad (10)$$

Proof. For a nonsingular symmetric form over $\mathbb{F}_2$, the characteristic functional $x \mapsto b(x, x)$ has a unique representing vector c . The standard orthogonal line/hyperbolic-plane decomposition shows $b(c, c) = n \bmod 2$. Since $\ell(x) = b(c, x)$ and $B_Q = b + \ell \otimes \ell$, its kernel equation is

$$B_Q(v, -) = 0 \iff v = \ell(v)c.$$

Thus B_Q is nonsingular when n is even; when n is odd its radical is $\langle c \rangle$, with $\ell(c) = 1$, and its restriction to $H = \ker \ell$ is nonsingular. Put $w = c$ in the odd case.

In even dimension, nondegeneracy supplies the unique a with $B_Q(a, -) = \ell$. Translation by a gives

$$Q(x + a) = Q(x) + Q(a) + \ell(x),$$

and hence $W(Q + \ell) = (-1)^{Q(a)}W(Q)$. The ordinary quadratic Gauss formula gives $W(Q) = (-1)^{\text{Arf}(Q)}2^{n/2}$. Substitution in (8) yields (9).

In odd dimension, pair the two cosets as h and $h + w$, $h \in H$. Since $w \in \text{rad } B_Q$,

$$\sum_{x \in V} i^{q(x)} = \sum_{h \in H} (-1)^{Q(h)} (1 + i(-1)^{Q(w)}).$$

The first factor is $(-1)^{\text{Arf}(Q|_H)}2^{(n-1)/2}$; the second has phase ζ_8 or ζ_8^7 . This proves (10). $\square$

Remark 4 (degenerate forms). Let $R = \text{rad } b$. Equation (2) gives $\ell|_R = 0$, hence $q|_R = 2Q|_R$. If $Q|_R \neq 0$, translation by a vector with $Q = 1$ cancels the full Gauss sum. If $Q|_R = 0$, the pair (ℓ, Q) descends to V/R and Theorem 3 applies to the nonsingular quotient. This is exactly the repository's `radical_anisotropic` boundary.

Remark 5 (the split is twisted under addition). Theorem 1 is a canonical bijection of data, not a direct-product law. Under the Brown–Wall orthogonal sum, if q_1, q_2 have coordinates (ℓ_1, Q_1) on V_1 and (ℓ_2, Q_2) on V_2 , then on $V_1 \oplus V_2$

$$\begin{aligned} \ell(x_1, x_2) &= \ell_1(x_1) + \ell_2(x_2), \\ Q(x_1, x_2) &= Q_1(x_1) + Q_2(x_2) + \ell_1(x_1)\ell_2(x_2). \end{aligned} \tag{11}$$

The same formula holds for pointwise addition on one source whenever that sum is considered. The last term is the binary carry from (4): it prevents this coordinate split from being a direct-product law and reflects the cited Brown–Wall $\mathbb{Z}/8$ Witt structure; it is not by itself a proof of that classification. A positionwise joint readout retains the carry and the correlation; separate marginal counts do not.

3 What the split means for play

Definition 6 (charge readout). A binary or four-class *charge readout* is a game rule whose complete play assigns a terminal element of $\mathbb{F}_2$ or $\mathbb{Z}/4$. It is weaker than a normal-play P/N realization: the terminal charge may be strategically forced without itself being the winner class.

Definition 7 (synchronized four-charge contract). Two charge readouts on the same input may be observed jointly and their ordered pair may be relabelled by a fixed bijection. Equivalence in this contract means equality of the resulting terminal labels for every input and complete play. It does *not* assert that the two game trees factor, that their strategies are independent, or that the pair is already one of the canonical outcome classes of a single normal-, misere-, or loopy-play game.

Proposition 8 (no independent over charge). **[PROVED]** *A four-class charge readout realizing a Brown form q is equivalent, by deterministic relabelling, to the joint readout (ℓ, Q) of Theorem 1. Conversely, joint readouts of ℓ and Q give a four-class readout by $(a, c) \mapsto \hat{a} + 2c$, provided the semantics admits synchronized products of two charge channels on the same input.*

Once ℓ is supplied, its coordinate is a solved local linear charge layer: after choosing any basis, attach a $$ component to precisely those active coordinates on which ℓ is one; their disjunctive sum has Grundy value $*\ell(x)$. All nonlinear dependence of the terminal charge on the input is therefore the ordinary quadratic readout Q . If one strengthens “charge readout” to “ P exactly when $Q = 0$ ”, the weighted-source Witt-FIFO theorem now supplies precisely that ordinary quadratic normal-play game. On the doubled slice $\ell = 0$, $q = 2Q$, this is the resolved Gold-form **tis** construction.*

Proof. The residue-to-bit map is the table in Corollary 2, and its inverse is $(a, c) \mapsto \hat{a} + 2c$; this is an observational equivalence of terminal labels, not a factorization theorem for game trees. For the linear realization, write $x = \sum_i x_i e_i$. The sum of the tagged $*$ components has Grundy coordinate $\sum_i x_i \ell(e_i) = \ell(x)$. Changing basis changes the presentation, not the resulting linear functional. The final sentence is the definition of the **tis** target applied to the ordinary quadratic form Q . $\square$

Remark 9 (the exact residual boundary). **[BOUNDARY]** Proposition 8 by itself does not turn a forced terminal charge into a normal-play winner. The later weighted-source matching theorem and phase-aware claim compiler do supply the Q bit as ordinary normal play. A synchronized ordered pair is still not automatically one canonical normal-, misere-, or loopy-play outcome of a single game; product closure is part of the four-charge contract. The point is narrower and complete: Brown’s four classes add a linear axis and a correlated second census, not a new algebraic kind of nonlinear charge/output problem.

4 The global value-level obstruction

The previous section concerns a selected finite $\mathbb{F}_2$ -space. A stronger naturality demand is that q belong to short-game values themselves rather than to one root-incomplete slice.

Definition 10 (ambient-coherent Brown value datum). For every finitely generated subgroup $M \leq \text{ShUg}$, choose

$$q_M : M \rightarrow \mathbb{Z}/4, \quad b_M : M \times M \rightarrow \mathbb{F}_2$$

satisfying (1), with b_M biadditive. The family is *ambient-coherent* if restriction along every inclusion $M \subseteq N$ preserves both functions. This is a generalized Brown-law datum on arbitrary game groups, deliberately stronger than a classical Brown form whose domain is elementary abelian.

Theorem 11 (short-game Brown collapse). **[PROVED]** *Every ambient-coherent Brown value datum on short games is zero:*

$$b_M = 0, \quad q_M = 0$$

for every finitely generated $M \leq \text{ShUg}$.

Proof. Moews proves

$$\text{ShUg} \cong \bigoplus_{\mathbb{N}} \mathbb{Z}[1/2] \oplus \bigoplus_{\mathbb{N}} (\mathbb{Z}[1/2]/\mathbb{Z}), \quad (12)$$

so multiplication by two, and hence by four, is surjective [2, Theorem 9].

Fix $x, y \in M$. Choose $u \in \text{ShUg}$ with $2u = x$ and enlarge M to the finitely generated subgroup $N = \langle M, u \rangle$. Coherence and bilinearity give

$$b_M(x, y) = b_N(2u, y) = 2b_N(u, y) = 0 \quad \text{in } \mathbb{F}_2.$$

Thus all polar corrections vanish and every q_M is additive. Choose $v \in \text{ShUg}$ with $4v = x$ and put $P = \langle M, v \rangle$. Then

$$q_M(x) = q_P(x) = q_P(4v) = 4q_P(v) = 0 \quad \text{in } \mathbb{Z}/4.$$

□

Corollary 12 (no exponent-two Brown quotient). **[PROVED]** *Every additive map from ShUg to an exponent-two group is zero; in particular $\text{ShUg}/2\text{ShUg} = 0$. Hence no nonzero $\mathbb{F}_2$ Brown module can be an additive quotient or homomorphic image of the full short-game group. This does not kill the intrinsic subgroup $\text{ShUg}[2]$: it contains $*$ and its root-incomplete finite subspaces support nonzero local Brown forms.*

Proof. If $f : \text{ShUg} \rightarrow A$ is additive, $2A = 0$, and $x = 2u$, then $f(x) = 2f(u) = 0$. □

This theorem is stronger than the earlier game-exterior obstruction on this specific target. It needs no Clifford algebra, coefficient embedding, or torsion hypothesis: the Brown polarization law and ambient roots suffice. The kernel-checked file `formal/0gdoad/BrownGame.lean` proves the abstract two-divisible/exponent-four implication and the $\mathbb{Z}/4$ specialization; Equation (12) remains an explicit source-pinned input rather than a Lean axiom.

5 Why a bare central extension does not supply the phase

The characteristic-two extraspecial construction does not lift verbatim from C_2 to C_4 . The obstruction is not commutativity but missing section data.

Proposition 13 (minimal framed cyclic model). **[PROVED]** *Consider the abelian exact sequence*

$$0 \longrightarrow C_4 \xrightarrow{a \mapsto 2a} C_8 \longrightarrow C_2 \longrightarrow 0.$$

Choosing $1 \in C_8$ as the lift of the nonzero quotient element gives the odd Brown line $q(1) = 1$; choosing $3 \in C_8$ gives $q(1) = 3$. Both have $b(1, 1) = 1$ and Brown phases 1 and 7, respectively. Therefore the bare extension determines neither q nor β .

Proof. Let $z = 2 \in C_8$ generate the embedded C_4 . For the first section, $2 \cdot 1 = z^1$ in additive notation, while for the second, $2 \cdot 3 = 6 = z^3$. In either case, substituting the nonzero vector twice in (1) forces $b(1, 1) = 1$. The Gauss sums are $1+i$ and $1-i$, hence the stated phases. More generally, changing a lift g to $z + g$ changes its double by $2z$; the square exponent is not fibre-invariant. □

The finite model and the non-fibre-invariance statement are also kernel-checked in `BrownGame.lean`. It corrects the tempting but false claim that a C_4 -central extension “corresponds exactly” to a Brown form. What corresponds to the full refinement is a suitably section-framed extension. Its commutator remembers only alternating information; the admissible section/framing carries the missing quadratic lift data, including the diagonal phase.

6 Disposition of `over`

The three readings from Section 1 now have exact answers.

- (1) **[PROVED]** On a selected finite $\mathbb{F}_2$ -space, the Brown four-class census is canonically the joint census of a locally realizable linear charge ℓ (once supplied) and one ordinary quadratic bit Q . Its complex imbalance is (7).
- (2) **[PROVED]** On short-game values with ambient coherence, every such Brown datum is zero by Theorem 11.
- (3) **[PROVED]** A bare C_4 -central extension is insufficient; a chosen section can encode the phase, but then the framing is supplied data.

Accordingly, under the explicit synchronized four-charge contract, there is no independent mod-8 charge/output problem left. The weighted-source Witt-FIFO theorem supplies the autonomous nonlinear normal-play bit, and crossing it with the linear bit gives the Brown classes automatically. This closes `over` as a separate invariant/output direction without asserting that the synchronized pair is already a single canonical normal/misere/loopy four-way outcome. If that single-game internalization is demanded, it is the exact residual boundary; it is not a surviving instance of the now-resolved binary `tis` problem.

References

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